

Executive summary

This report presents both parts of the forecasting assessment using quarterly time series from the M3 competition. Part 1 focuses on a single series, ID 1357, while Part 2 extends the analysis to 100 quarterly series with IDs 1001–1100. The report therefore evaluates both a detailed, manually specified forecasting exercise and a larger-scale batch-forecasting exercise.

In Part 1, series 1357 was explored using time-series plots, descriptive statistics, seasonal summaries, decomposition, autocorrelation analysis and stationarity tests. The analysis indicated a non-stationary series with a downward trend and quarterly seasonal behaviour. Three models were then selected intuitively: a regression model containing trend and seasonal components, an additive ETS model and a manually specified seasonal ARIMA model. Residual diagnostics, forecasts and 80% and 95% prediction intervals were produced for each model. On the eight-quarter holdout sample, the regression model achieved the lowest RMSE, although its residuals showed evidence of remaining autocorrelation, meaning that its strong test-period performance should be interpreted alongside its diagnostic limitations.

Part 2 applied a batch-forecasting approach to 100 quarterly M3 series. Automatic ETS and ARIMA models were fitted, and their forecasts were combined using equal weights. Naïve and seasonal naïve forecasts were included as benchmark methods. Forecast accuracy was evaluated using the mean and median of Mean Absolute Scaled Error (MASE) and symmetric Mean Absolute Percentage Error (sMAPE). These measures were selected because the series can differ in scale and magnitude, making scale-independent accuracy measures appropriate for cross-series comparison.

The batch analysis showed substantial variation in the structural characteristics of the series. According to the automatic ETS classifications, most series contained a trend component, while fewer contained seasonality. The automatic ARIMA procedure generally selected parsimonious models, with low autoregressive and moving-average orders common across the dataset and non-seasonal differencing frequently required. ETS won the lowest error for more individual series than the combined model. Nevertheless, the combined ETS–ARIMA model produced marginally lower mean and median MASE and sMAPE values. Because these differences were small, the results do not demonstrate a practically meaningful superiority of the combined model.

Overall, the findings support using different approaches according to the forecasting context. For a single series, detailed exploratory analysis and residual diagnostics can support an interpretable, problem-specific model choice, as demonstrated in Part 1. For a larger collection of series, automatic methods provide an efficient and scalable alternative. ETS is recommended as the default batch-forecasting method because it won the greatest number of individual series and is simpler to interpret and implement, while the combined model remains a competitive alternative where a small improvement in aggregate accuracy justifies additional complexity. The results also suggest that selecting the best model separately for each series may be preferable to applying one method universally.

Main Report

Introduction

Forecasting plays a central role in business decision-making because organisations routinely need estimates of future demand, sales, costs and resource requirements before those outcomes are observed. In practice, decisions on inventory, staffing, budgeting, procurement and capacity planning are often made under uncertainty, so forecast quality affects both operational efficiency and strategic performance. Poor forecasts can lead to stockouts, excess inventory, underused capacity, weak cash-flow planning and avoidable cost, which is why forecasting methods should be evaluated systematically rather than selected on intuition alone.

This report examines forecasting from two complementary perspectives. Part 1 focuses on a single quarterly time series from the M3 competition with ID 1357, and develops three models through detailed exploratory analysis and intuitive model selection: a regression model with trend and seasonal effects, an exponential smoothing model from the ETS family and a seasonal ARIMA model. The purpose of this first part is to demonstrate how a forecaster can study the structure of one series in depth, identify plausible dynamics and assess competing models using residual diagnostics, forecasts and out-of-sample performance.

Part 2 extends the analysis to a batch-forecasting setting involving 100 quarterly M3 series with IDs 1001 to 1100. In this part, forecasting methods must be applied consistently across a larger collection of series with potentially different characteristics, including differences in trend, seasonality, volatility and available training history. To address this, the report evaluates automatically selected ETS and ARIMA models, together with an equal-weight combination of the two, and compares them with naïve and seasonal naïve benchmark methods.

The distinction between the two parts is important from a business analytics perspective. A manually selected model may be useful when a decision-maker is concerned with one critical series and can justify time spent on detailed diagnostics. By contrast, large organisations often need forecasts for many products, regions or business units simultaneously, making scalable and automated forecasting procedures more relevant. This report therefore considers both the interpretability of carefully selected models for a single series and the practicality of automatic forecasting methods for multiple series.

The report has three main aims. First, it evaluates whether the three intuitively selected models in Part 1 can adequately represent the dynamics of series 1357 and produce accurate eight-quarter-ahead forecasts. Second, it assesses the forecasting performance of ETS, ARIMA and combined forecasts across 100 quarterly series in Part 2 using scale-independent error measures. Third, it compares model performance not only in terms of average accuracy, but also in terms of consistency across series and practical usefulness for managerial decision-making.

To achieve these aims, the report combines exploratory time-series analysis, decomposition, stationarity assessment, residual diagnostics and out-of-sample forecast evaluation. In Part 1, the three selected models are compared using forecast plots, prediction intervals and Root Mean Squared Error on the holdout sample. In Part 2, the methods are evaluated using Mean Absolute Scaled Error and symmetric Mean Absolute Percentage Error, allowing comparisons across series with different scales and magnitudes. Benchmark forecasts are also included so

that the value of the proposed approaches can be judged against simple alternatives rather than in isolation.

The report is structured accordingly. It first introduces the data and exploratory analysis, then presents the modelling approaches used in each part, followed by the evaluation framework and empirical results. The final sections discuss the findings, limitations and managerial implications, with the goal of identifying forecasting approaches that are not only statistically credible but also practically relevant in business settings.

Data and Exploratory Analysis

In Part 1 only one series was examined, identified by ID 1357. The reason for focusing on one series was to investigate the characteristics of the data and manually choose the parameters of 3 forecasting models. The training data consisted of 62 quarterly observations, beginning in the 2nd quarter of 1978 and ending in the 3rd quarter of 1993. The following eight observations were the out-of-sample test period.

Part 2 involves the examination of 100 quarterly times series, with IDs from 1001 to 1100. Similarly to Part 1, these are quarterly series and the 8 quarter forecast horizon is considered for the evaluation of the models. However, the models' parameters are not intuitively selected based on the characteristics of each series. Instead, automatic model creation was carried out using the "auto.arima()" and "ets()" of the "forecast" package in R.

The main difference between the 2 parts is that Part 1 focuses on detailed forecasting for an important individual data series and Part 2 emphasises forecasting for many data series with the purpose of identifying a better forecasting method for series with similar characteristics.

Part 1: Series 1357

Firstly, the data series was plotted :

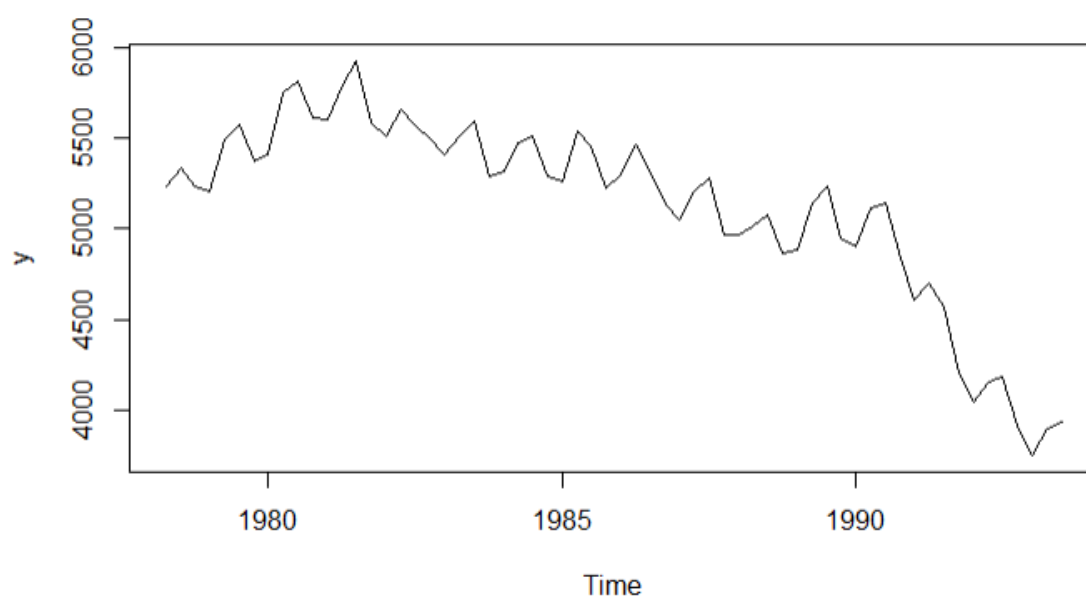


Figure 1: Time plot of series 1357

The series shows a generally declining trend over the sample period, along with quarterly fluctuations. The series had periods of temporary increases and decreases, with a more pronounced decline towards the end of the training sample. This indicates that a forecasting model should be able to account for both long term movement and repeated within year patterns.

The 8 observations in the test data represent two future years. This is useful as the forecast evaluation covers two repetitions of the quarterly seasonal cycle rather than one single future quarter.

Descriptive statistics were calculated using the “summary()” and “sd()” functions in R. The statistics are listed below :

Min	3749
1 st Quartile	4949
Median	5252
3 rd Quartile	5498
Max	5927
Mean	5127
Standard Deviation	520

While these statistics provide information about the location and dispersion of the series, interpretation must be carried out along with the time plot, as these statistics do not preserve timing information. The time plot indicates that the variation was not constant throughout the sample, as the level of the series changed over time and later observations showed greater decline than earlier observations. Therefore, standard deviation should not be interpreted as evidence that the series has constant volatility.

Quarterly distributions were examined through a boxplot of values grouped by each quarter.

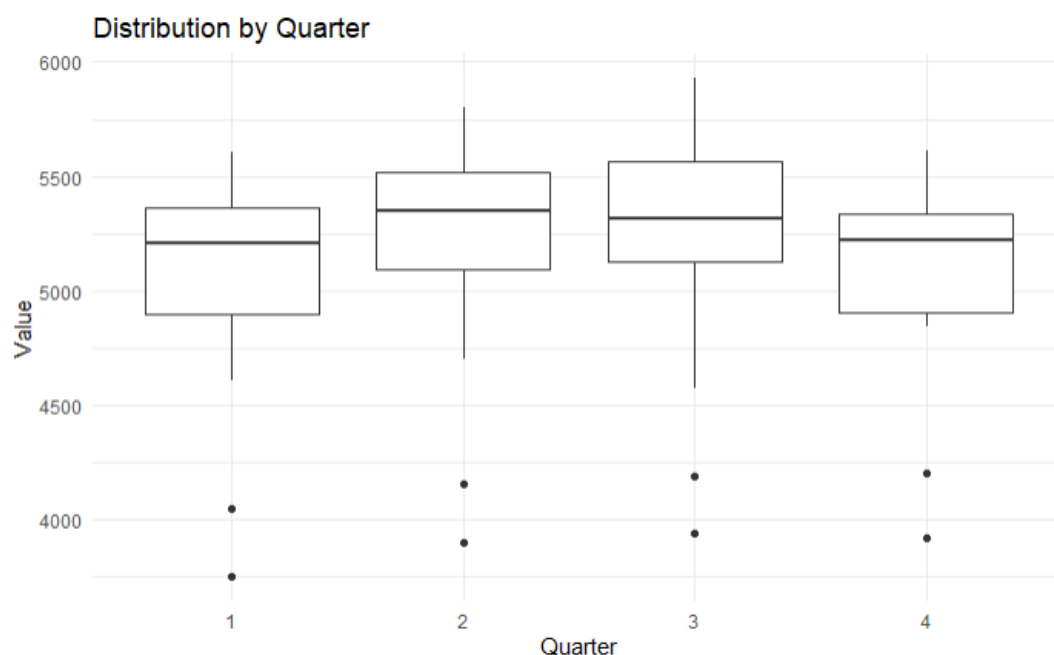


Figure 2: Box plot of series 1357 separated by quarter

The results support the presence of quarterly seasonal behaviour. While there are 2 points for each quarter characterized as outliers, they should still be considered in our modelling, as they refer to the last 2 years of observations where there was a greater decline.

Dependence and Stationarity

Autocorrelation (ACF) and partial autocorrelation (PACF) plots were used to examine the temporal dependence.

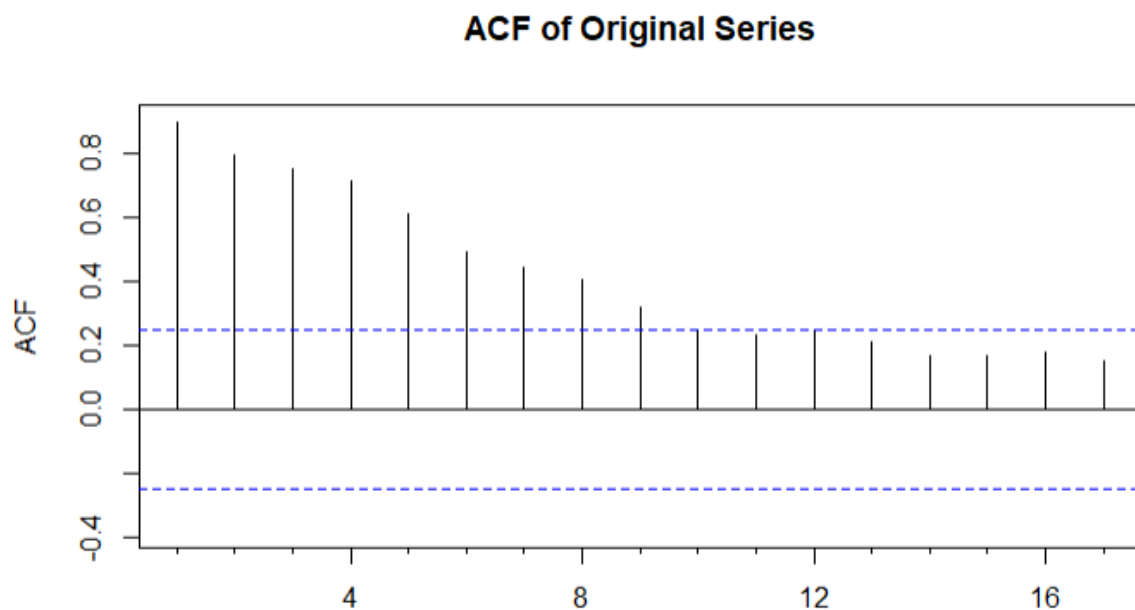


Figure 3: Autocorrelation Function plot of original series

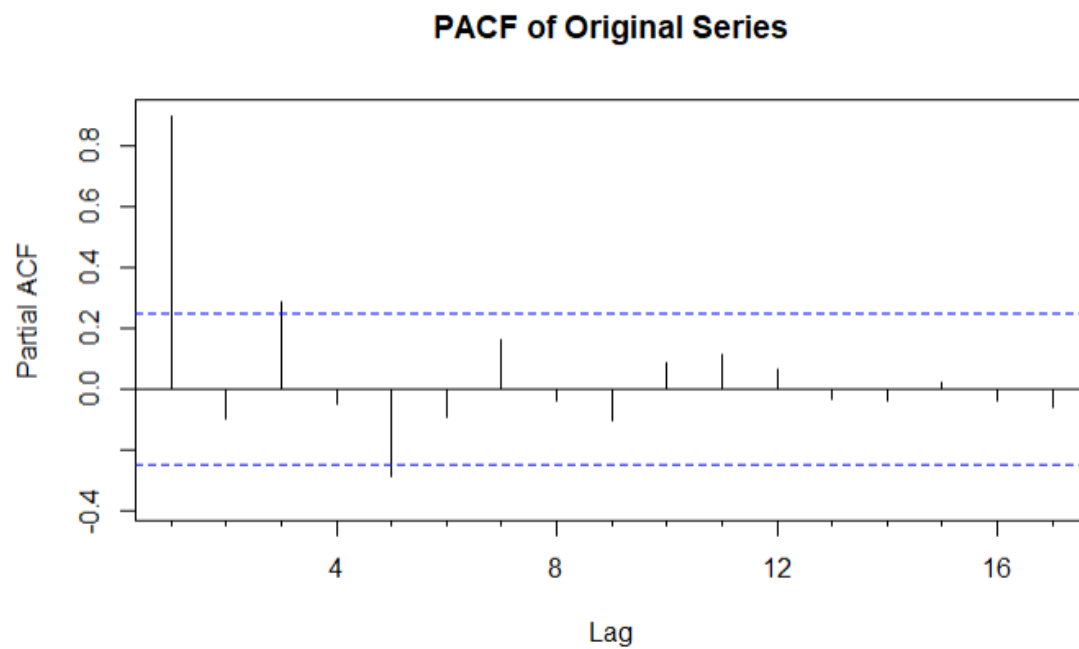


Figure 4: Partial Autocorrelation Function plot of original series

The ACF showed significant spikes for several lags, which indicates that recent observations were dependent on previous observations. This was consistent with the trend and seasonality displayed in Figure 1. The PACF was also examined to spot possible lag structures that could lead us to a better ARIMA specification.

To remove the non-stationarity of the series, differencing was carried out. Firstly, first differencing led to the following ACF and PACF plots:

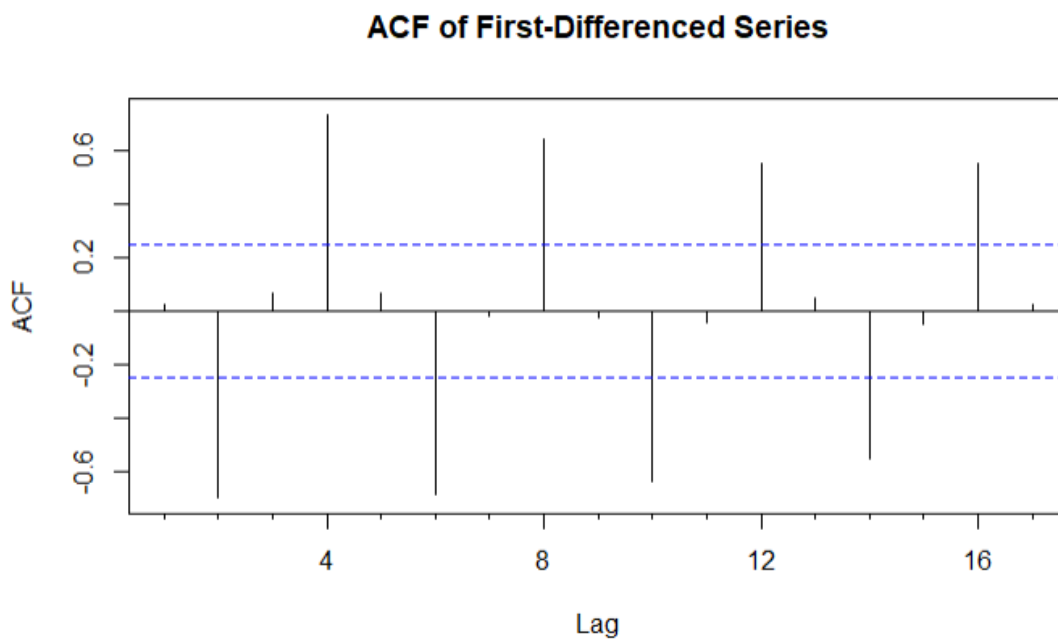


Figure 5: ACF of First-Differenced series

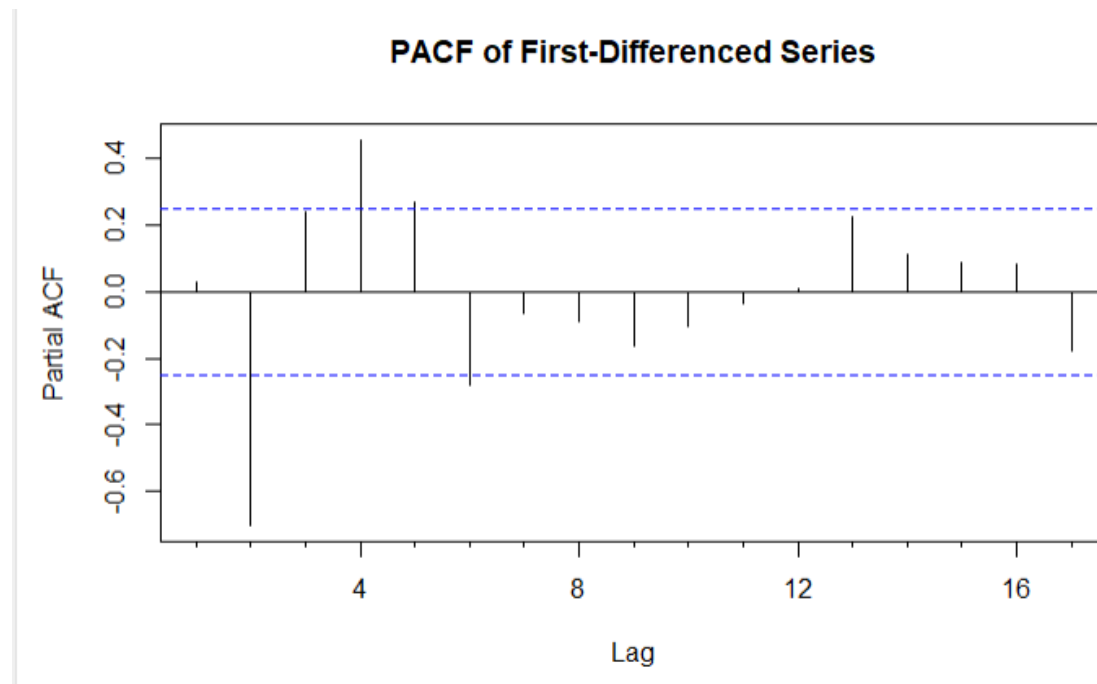


Figure 6: PACF of First-Differenced series

In figure 5 we can clearly see significant spikes every 4 lags – which is the seasonality of the quarterly data – therefore seasonal differencing at lag 4 was also considered to stabilize the transformed series.

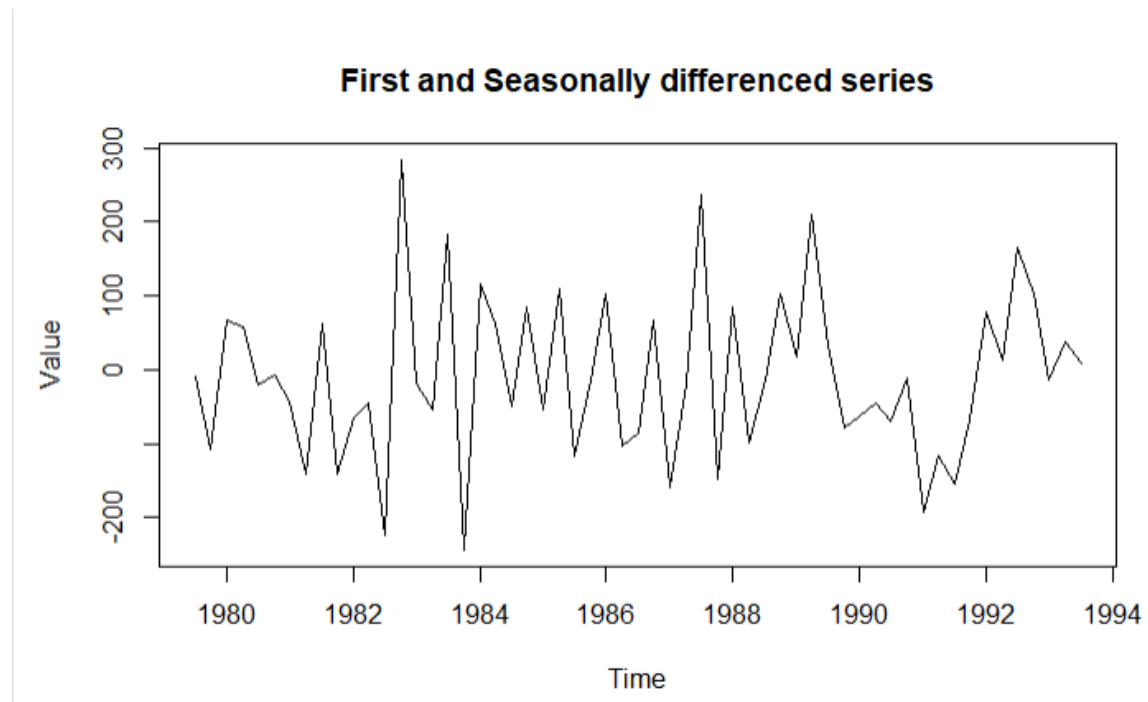


Figure 7: Values of First and Seasonally differenced series

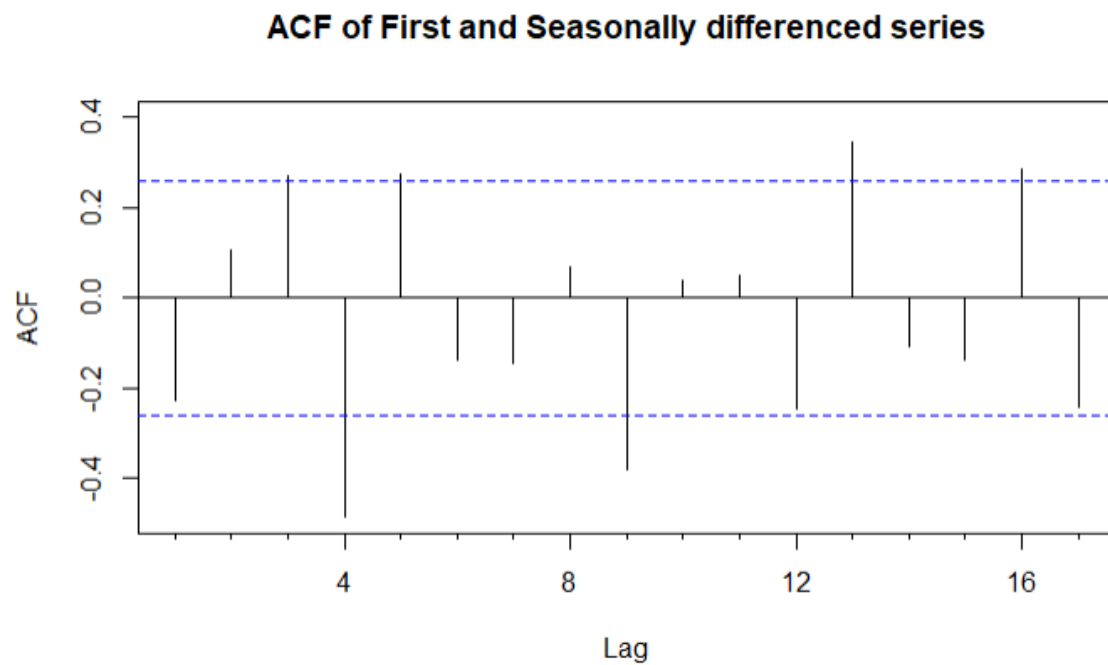


Figure 8: ACF of First and Seasonally differenced series

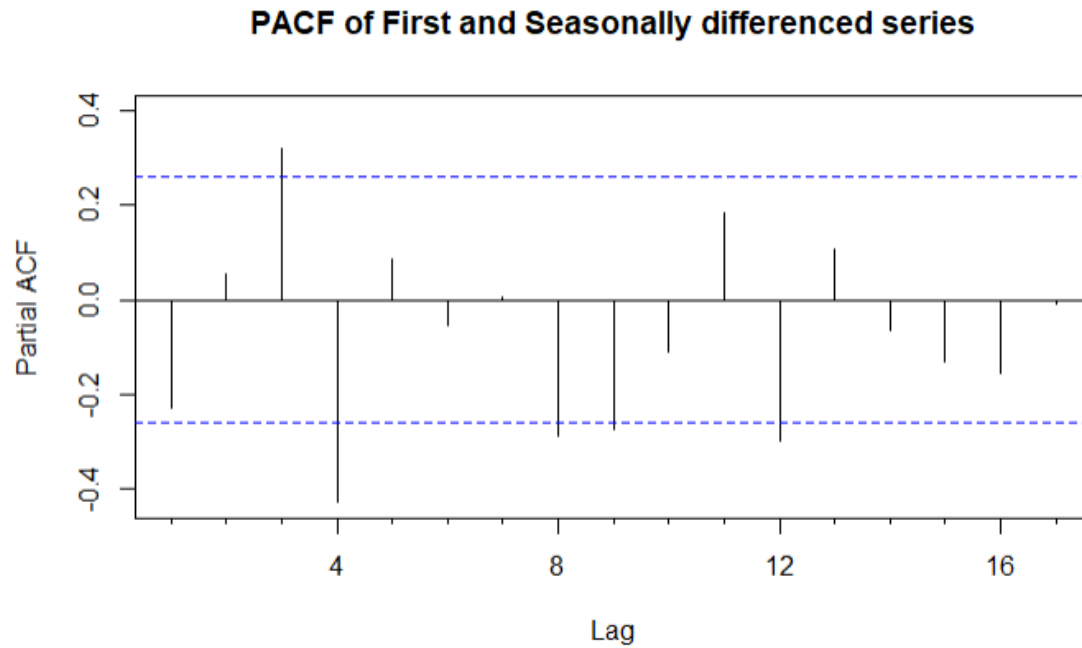


Figure 9: PACF of First and Seasonally differenced series

After applying both first and seasonal differencing to the series, it appeared more stable with the slow autocorrelation decline substantially reduced. There is still some significant spikes at ACF and PACF suggesting some seasonal dependence.

The stationarity of the original series was tested through the Augmented Dickey-Fuller (ADF) and KPSS tests. The ADF test did not provide any evidence against the non-stationarity of the series and KPSS confirmed that conclusion.

Decomposition

Classical additive and multiplicative decompositions, with STL decomposition, were used to identify the seasonal, trend and remainder components.

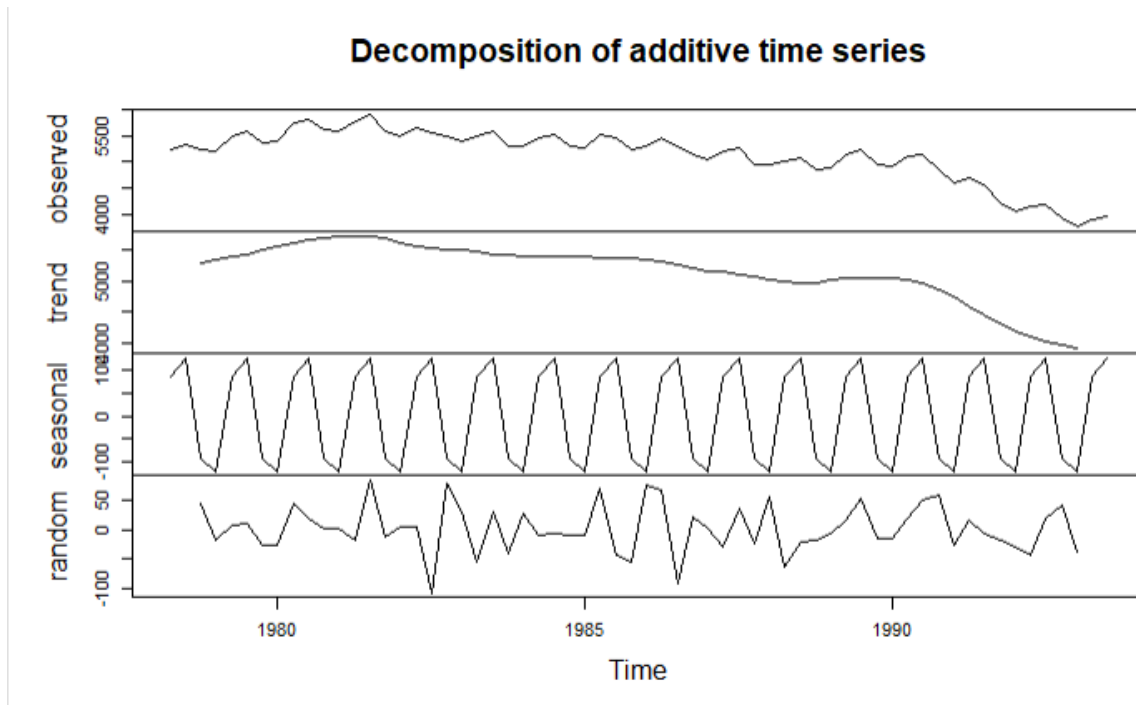


Figure 10: Classical Additive decomposition of series

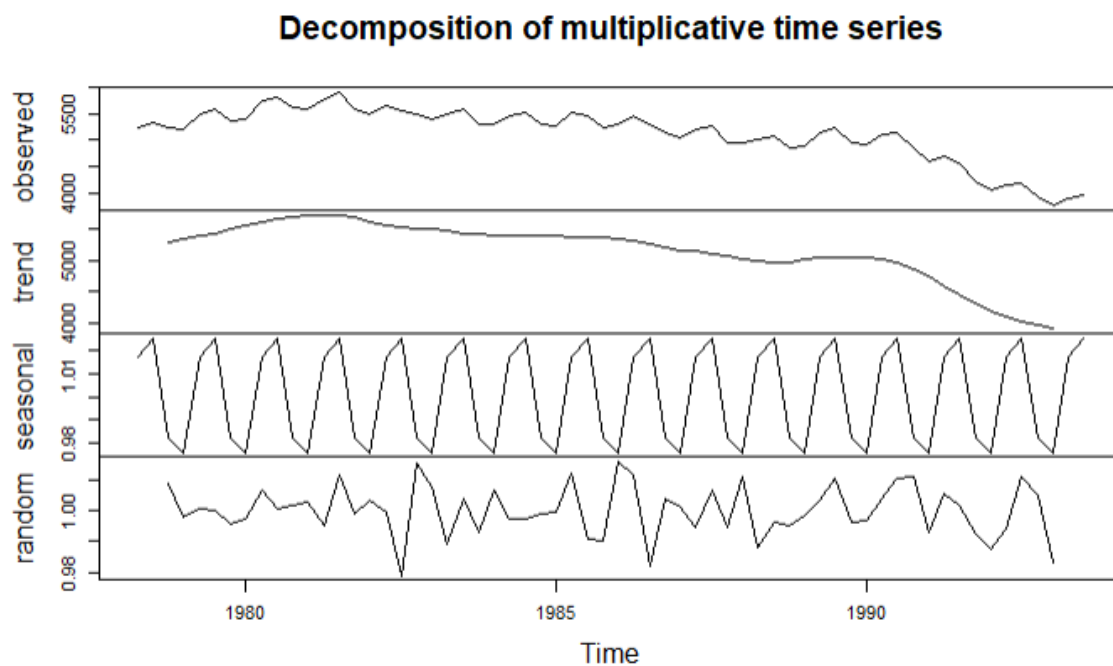


Figure 11: Classical Multiplicative decomposition of series

The seasonal component in additive decomposition seems to be relatively stable compared with the level of the series, suggesting that an additive seasonal model may suffice.

The above findings help us conclude the following :

- The regression model should include a trend and seasonal term
- The ETS model should have additive seasonality

- The ARIMA model should have seasonal components due to the autocorrelation at quarterly lags

Part 2: Batch Data

For the analysis of the 100 series with IDs from 1001 to 1100, a summary table was first created including information on the period, type, training length, test length and frequency. This showed that all series have quarterly frequency and an eight-quarter forecast horizon, but the training data differs in length. While most series contained 44 training observations, five had 26 and one had 16. A series with 44 observations (11 years in this case) provides more information for estimating trend, seasonal effects and serial dependence than a series with only 16 observations (4 years). Shorter history may also allow random variation to affect the conclusions on seasonal behaviours of the data.

The concentration of series at the common training length of 44 demonstrates that the dataset is comparable, but the shorter series should not be ignored. In the real world, this issue would correspond to products/entities with different length of sales history.

Four random series were then plotted to provide examples of variation present within the dataset. The series demonstrate that there can be differences in their levels, trends and seasonal patterns. This variation encourages automatic model selection, since manually choosing one model to be fitted to all series – with their different characteristics – would not be pragmatic and its evaluation would not be interpretable.

Trend and Seasonality

An ETS model was fitted to each series to examine the structural differences within the batch of series using the “components[x]” attribute of the fitted model in R. According to the automatic ETS model components, 77 out of the 100 series had a trend component and 25 out of the 100 had a seasonal component.

This classification should be interpreted as a description of the components selected by the automatic ETS procedure and not as an assumption that the data-generating process is such i.e. the actual nature of the data. An example that demonstrates this difference would be a series with weak seasonality that is not selected by the information-criterion-based model (as in the automatic modelling the goal is to minimise the information criteria).

ETS can adapt its error, trend and seasonal components to individual series, while ARIMA can deal with serial dependence and non-stationarity by differencing and autoregressive or moving average terms. A combined model allows those 2 different methods to contribute simultaneously to forecasting.

The issue of structural differences in the data (scale and training length) encouraged the use of automatic ETS and ARIMA models. Then an equal weights combination of the above methods was used – where each method contributes equally to the final forecast – along with 2 benchmark methods, namely Naïve and Seasonal Naïve.

Forecasting Methodology

Part 1

In this part, one Regression, ETS and ARIMA models were manually created based on the exploratory analysis which indicated a downward trend, quarterly seasonality and non-stationarity.

Regression model

The trend and seasonality were used as predictors. The trend term captures the long-term while seasonal terms allow differences between quarters. Besides the y-intercept, the model has 3 parameters to optimize for each quarter except for the Q1 which acts as a reference value. Such a model assumes that the trend and seasonality remain relatively stable, which may not be optimal.

ETS model

An ETS with additive error, trend and seasonality was created. This specification was informed by the seasonal fluctuations appearing approximately constant in the exploratory analysis. The trend was not damped as no strong evidence was provided to do so.

ETS models are advantageous compared to regression models, as the estimated level, trend and seasonal components are updated as more observations become available, which makes them more dynamic.

ARIMA model

The ARIMA model was specified as $ARIMA(0,1,0)(1,1,0)_4$. First regular differencing was selected to address the non-stationarity of the series, while seasonal differencing at lag-4 for the quarterly repetition.

The trend component captured the downward movement, while the seasonal component represented recurring quarterly variation. The seasonal autoregressive term captures dependence between observations separated by 1 year (4 quarterly observations). This specification was inspired by the ACF and PACF plots of the differenced series.

Part 2

The first proposed model was an automatic ETS for each of the 100 series. This is done by selecting appropriate trend, error and seasonal components to minimize an error.

Then, automatic ARIMA models were created for each time series. Each model aims to reduce an information criterion. Differencing, autoregressive and moving average orders differ between each series.

The 3rd proposed method was an equal weight combination of the ETS and ARIMA. This is done in the following way:

$$y' = 0.5y'_{ets} + 0.5y'_{ARIMA}$$

y' – prediction of the combined model

y'_{ets} – prediction of the automatic ETS

y'_{ARIMA} – prediction of the ARIMA model

Then 2 models were created for each series acting as benchmarks : a naïve and a seasonal naïve model. The naïve model forecasts the next period using the last observed value, while the seasonal naïve used the last observed value for the same period (in this case previous year's observation on the same quarter).

Benchmark models are usually simpler and inform us on whether or not complicating a model adds any forecasting value.

Evaluation Measures

In Part 1, the three manually selected models for series 1357 were compared using Root Mean Squared Error (RMSE) on 8 out-of-sample observations. RMSE is appropriate for penalising large errors while it also keeps the original units of the data, which make the errors comparable with the scale of the series.

In Part 2, the analysis was done in 100 series, making scale-independent measure necessary. Mean Absolute Scaled Error (MASE) was selected as it compares forecast accuracy relative to a benchmark error and allows results to be compared across series with different magnitudes. The scaling factor was based on seasonal differences at lag-4 (since the data is quarterly). Values below 1 indicate that a method perform better than the seasonal benchmark.

Symmetric Mean Absolute Percentage Error (sMAPE) was also used as a second measure. It is scaled-independent as it expresses forecast error relative to the combined magnitude of the actual and forecast values, providing percentage-based accuracy measures.

Both the mean and median values of MASE and sMAPE were reported across the 100 series. The mean summarises overall performance while the median provides a measure of the typical series.

Results

Part 1

The regression model with trend and seasonal terms captured the broad downward trend of the series and quarterly pattern, but the residual diagnostics showed that it did not fully account for the serial dependence in the data.

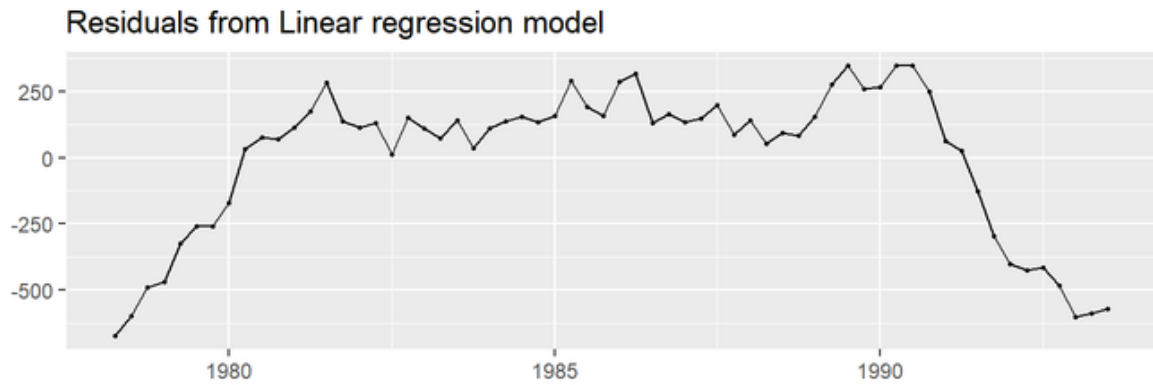


Figure 12: Regression Residuals plot

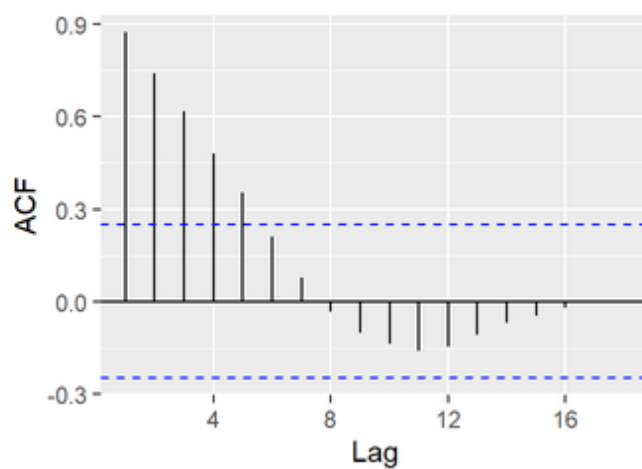


Figure 13: Regression ACF plot of residuals

The residual plot showed long runs of positive and negative errors, and the ACF indicates residual remaining autocorrelation. The Ljung-Box and Shapiro-Wilk tests both rejected their null hypotheses, suggesting dependence and non-normality of errors.

The additive ETS model had an improved residual pattern compared to the regression model.

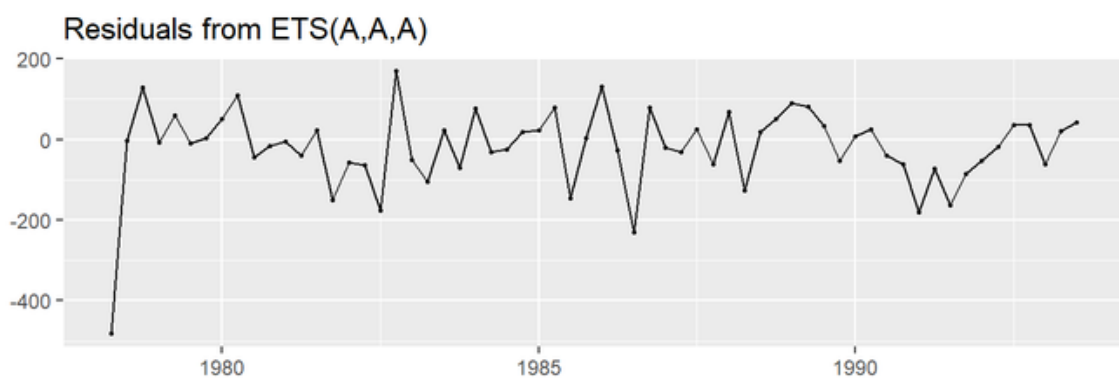


Figure 14: ETS Residuals plot

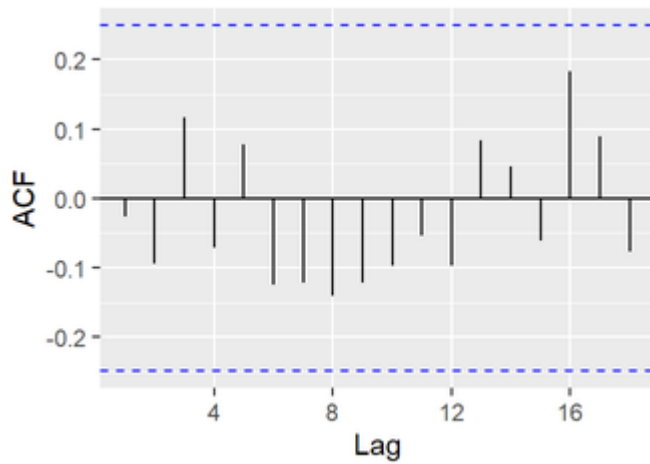


Figure 15: ETS ACF plot of residuals

Residuals fluctuated more closely around zero and the ACF did not show statistically significant spikes. The Ljung-Box test suggested that there is no remaining autocorrelation, however, the Shapiro-Wil test indicated that the residuals were not normally distributed.

The seasonal ARIMA model produced the strongest residual diagnostics among the three models.

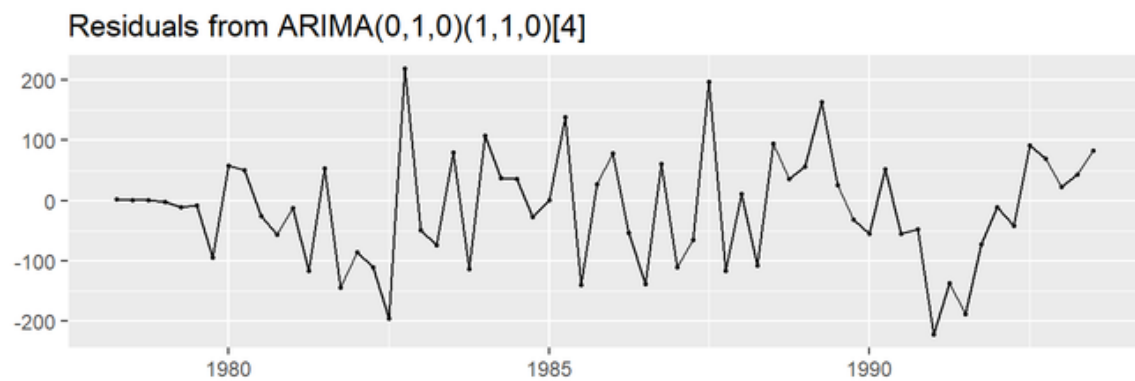


Figure 16: ARIMA Residuals plot

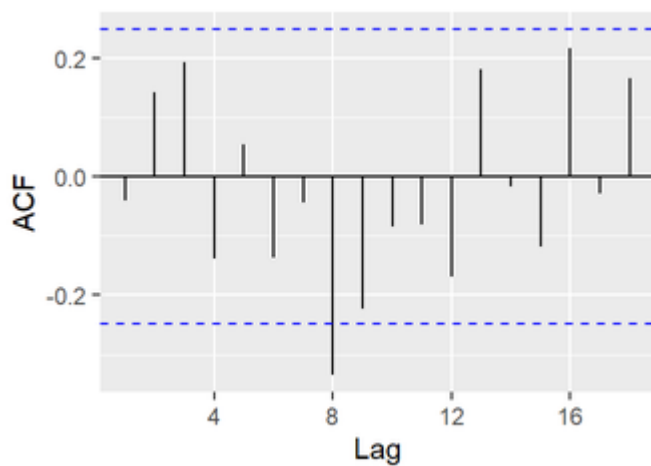


Figure 17: ARIMA ACF plot of residuals

The fitted model had no strong evidence of residual autocorrelation and the residuals were closer to normality than in the other models. This suggests that the ARIMA model captured the temporal dependence in the series more effectively than the regression and ETS alternatives.

However, different conclusions can be drawn from the out-of-sample forecast of the next 8 quarters. The regression model achieved the lowest RMSE at approximately 437, compared with about 512 for the ETS and 566 for the ARIMA model. Regression produced the most accurate forecasts despite its residual diagnostics. These results tell us that for this series and this test period specifically, the continuation of the estimated linear trend and seasonal structure was more effective for forecasting than more dynamic modelling.

Part 2

From the ETS classifications, 77 series included a trend component and 25 a seasonal component. The automatic ARIMA models preferred generally low autoregressive and moving average orders. Non-seasonal differencing was selected frequently and seasonal ARIMA terms were less widespread than we would expect from a quarterly data.

The summary table of the mean and median errors of the 5 methods used showed that the combined ETS-ARIMA model achieved the lowest mean and median values for both MASE and sMAPE, but only by a small margin. This indicates that it performed slightly better in terms of average accuracy and typical series outcome.

However, ETS achieved the lowest error for more individual series than the combined model. The win-count results show that the ETS model was more often the best-performing method at the level of individual series. This shows us that the most frequent winner is not necessarily the one with the lowest overall mean or median error.

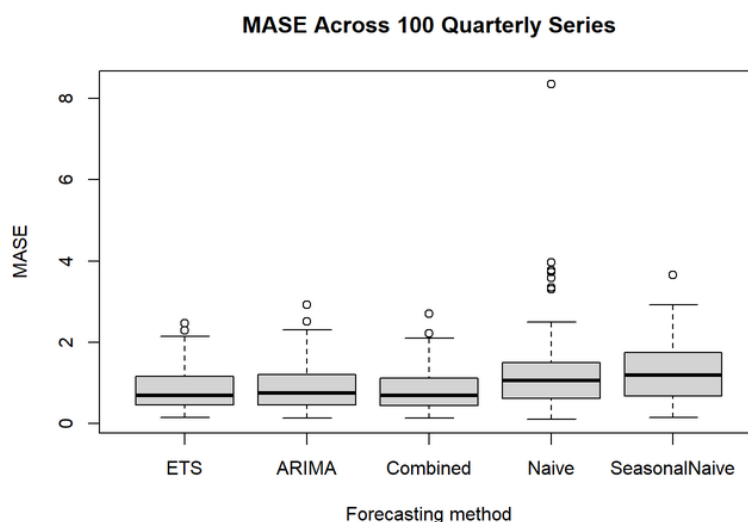


Figure 18: MASE Box-plot of all methods for 100 Quarterly Series

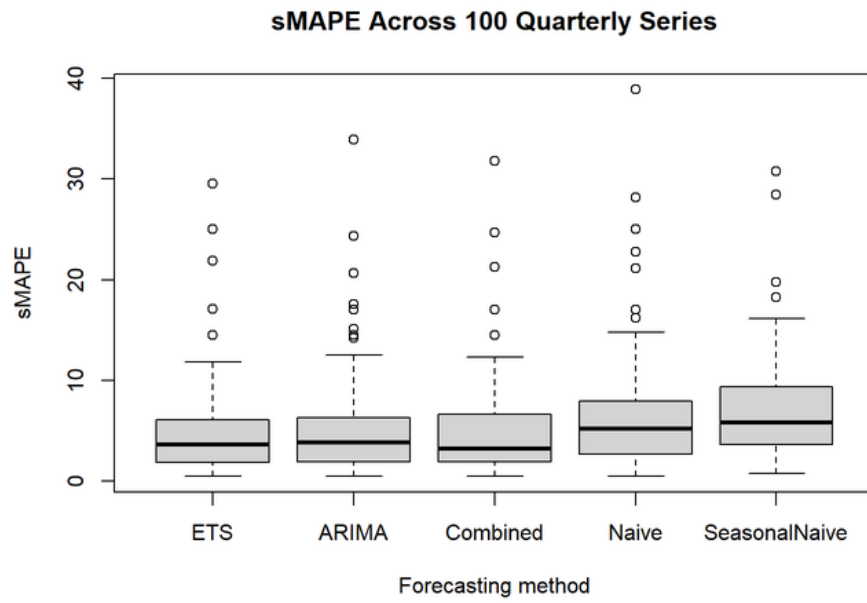


Figure 19: sMAPE Box-plot of all methods for 100 Quarterly Series

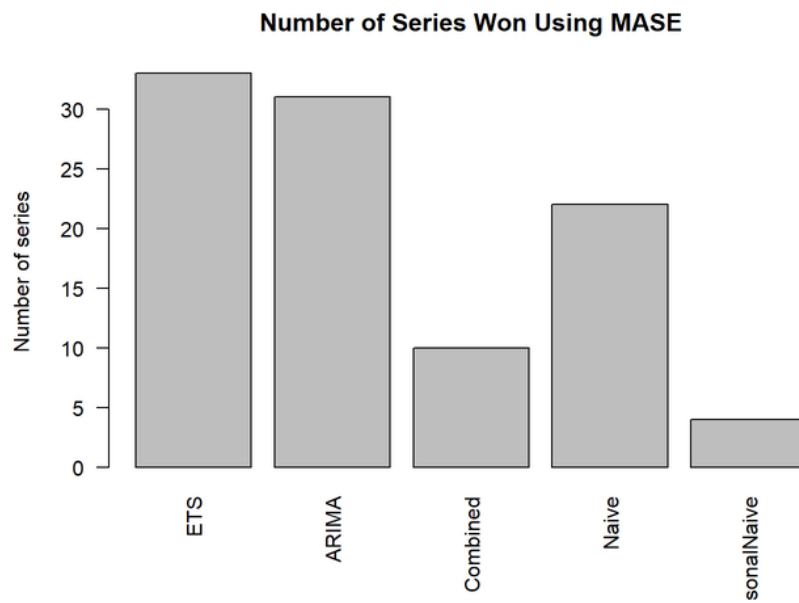


Figure 20: Number of wins per method based on MASE

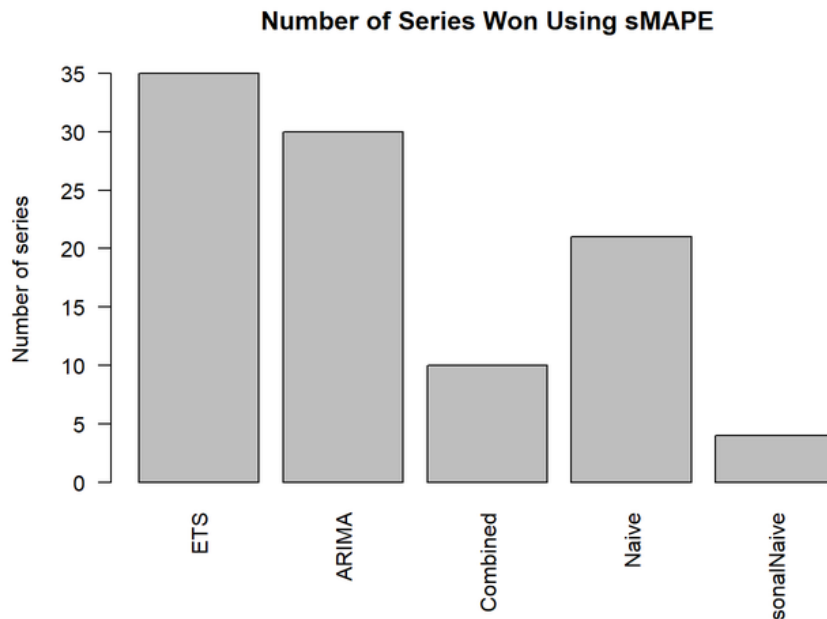


Figure 21: Number of wins per method based on sMAPE

The proposed methods were competitive with the simple benchmarks alternatives. Among the proposed methods, ARIMA was performing worse than ETS and the combination when judged by the summary of error measures, while the combined model and ETS were the 2 strongest candidates overall.

The box-plot shows that the combined model had slightly lower central tendency, while ETS remained competitive and achieved more individual wins.

Discussion, Limitations and Managerial Implications

From both parts of the assignment we can conclude that the model that appears strongest according to one criterion is not always the best according to another.

In Part 1, the regression model produced the lowest RMSE on the forecast horizon, even though its residual diagnostics were weaker than those of the ETS and ARIMA models. The ARIMA model produced the strongest residual diagnostics but had the least accurate forecasts for the next 8 quarters. This shows that in-sample performance does not always agree with out-of-sample performance. For practical forecasting, model adequacy and forecast accuracy therefore need to be assessed together rather than independently.

Part 2 leads to similar conclusion but at different scales. The combined ETS-ARIMA forecasts produced slightly lower mean and median MASE and sMAPE, suggesting a small advantage in overall and typical accuracy over 100 series. However, ETS had the lowest error more often than the combined model. As the aggregate differences were only marginal, there is no strong evidence to show that the combined model is superior.

Forecast combinations are often found to improve robustness as they do not rely on one potentially misfitting method. Equal-weight combinations are commonly used because they are transparent and can work well when there is insufficient information to estimate optimal weights.

From a managerial perspective, ETS seems like a suitable default method as it is relatively simple, easy to interpret and had a greater win-score. Additionally, detailed exploratory analysis is valuable when forecasting an important individual series.

There are however some limitations to be acknowledged. Part 1 used only one series and one eight-quarter holdout period, while part 2 also relied on a single fixed test period. Part 2 included 6 series with different training lengths than the typical length of 44 quarters. Also, the results from Part 1 and Part 2 should carefully be compared, as they are using different error measurements.